

Topic IV: Earth Mover's Distance

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1 Earth Mover's Distance

Many types of data can be described naturally as probability distributions. For example, gray-scale images with non-negative values can be normalized so their total mass is 1; the bag-of-words model in textual analysis treats each document as a distribution over keywords; and in the theory of diffusion maps [2] any individual node in a graph can be assigned a probability distribution over all the nodes. A natural question, then, is to find good metrics for comparing probability distributions.

The Earth Mover's Distance (in the discrete setting) is defined as follows. Suppose $\mathcal{X}$ is a finite set of size n . Given two probability vectors $\mathbf{p}$ and $\mathbf{q}$ of length n , that is, two vectors satisfying $\mathbf{p} \geq 0$, $\mathbf{q} \geq 0$, and

$$\sum_{i=1}^n p_i = \sum_{i=1}^n q_i = 1, \quad (1)$$

we define a *transportation plan* from $\mathbf{p}$ and $\mathbf{q}$ to be a matrix $\mathbf{\Pi} = (\pi_{ij})$ with non-negative entries satisfying

$$\sum_{j=1}^n \pi_{ij} = p_i \quad (2)$$

for all $1 \leq i \leq n$, and

$$\sum_{i=1}^n \pi_{ij} = q_j \quad (3)$$

for all $1 \leq j \leq n$. We interpret π_{ij} as an amount of mass shipped from i to j ; the constraints tell us that the total mass shipped out of any location i must be equal to p_i , and the total amount of mass shipped into location j must be equal to q_j ; that is, $\mathbf{\Pi}$ describes a shipment plan for transforming the distribution $\mathbf{p}$ to the distribution $\mathbf{q}$. Denote by $\mathcal{P}(\mathbf{p}, \mathbf{q})$ the set of all transportation plans from $\mathbf{p}$ to $\mathbf{q}$.

Suppose we specify an n -by- n distance matrix $\mathbf{D} = (d_{ij})$, satisfying $d_{ij} \geq 0$, $d_{ij} = d_{ji}$, and $d_{ik} \leq d_{ij} + d_{jk}$. We interpret d_{ij} as the cost of moving a unit of mass from the i -th point in $\mathcal{X}$ to the j -th point in $\mathcal{X}$. We then define the *cost* of a transportation plan $\mathbf{\Pi}$ in $\mathcal{P}(\mathbf{p}, \mathbf{q})$ to be

$$\text{cost}(\mathbf{\Pi}) = \sum_{i=1}^n \sum_{j=1}^n \pi_{ij} d_{ij} = \langle \mathbf{\Pi}, \mathbf{D} \rangle \quad (4)$$

That is, if it costs $m \cdot d_{ij}$ to move mass m from location i to location j , then $\text{cost}(\mathbf{\Pi})$ is the total cost of executing the transportation plan described by $\mathbf{\Pi}$.

The *Earth Mover's Distance (EMD)* between $\mathbf{p}$ and $\mathbf{q}$ is defined as follows:

$$\text{EMD}(\mathbf{p}, \mathbf{q}) = \min_{\mathbf{\Pi} \in \mathcal{P}(\mathbf{p}, \mathbf{q})} \text{cost}(\mathbf{\Pi}). \quad (5)$$

That is, $\text{EMD}(\mathbf{p}, \mathbf{q})$ is the cheapest way of transporting mass distributed as $\mathbf{p}$ to mass distributed as $\mathbf{q}$.

EMD is often a natural way of comparing distributions, because it can exploit the underlying geometry on $\mathcal{X}$ as defined by the distance $\mathbf{D}$. Next, we will look at simple examples of EMD in special cases where it can be computed analytically. However, this is unfortunately not the typical case, and indeed one of the disadvantages of EMD is that evaluating it is usually costly, on the order of $\mathcal{O}(n^3 \log(n))$ operations. If the goal is to compute distances between pairs of distributions in a large data set, this can be prohibitively expensive. However, in many data science applications, finding the optimal transport between $\mathbf{p}$ and $\mathbf{q}$ is not critical; rather, one only wants to compute the distance $\text{EMD}(\mathbf{p}, \mathbf{q})$ itself, or an approximation to it. It turns out that there are fast approximations to EMD, which we will explore later.

Remark 1. The definition of EMD still makes sense if the entries of $\mathbf{D}$ do not satisfy the triangle inequality (that is, the cost function need not be a proper distance). Furthermore, the distributions $\mathbf{p}$ and $\mathbf{q}$ can be supported on different sets $\mathcal{X}$ and $\mathcal{Y}$, so long as there is a cost defined of moving mass from each point in $\mathcal{X}$ to each point in $\mathcal{Y}$; in this case, $\mathcal{X}$ and $\mathcal{Y}$ need not have the same size.

1.1 General properties of EMD

We state some general properties of EMD. First, $\text{EMD}(\mathbf{p}, \mathbf{q})$ defines a proper distance between probability distributions:

Theorem 1.1. *Let $\mathcal{X}$ be a finite set with metric d . Then:*

1. *For all probability distributions $\mathbf{p}$ and $\mathbf{q}$ over $\mathcal{X}$, $\text{EMD}(\mathbf{p}, \mathbf{q}) \geq 0$, and $\text{EMD}(\mathbf{p}, \mathbf{q}) = 0$ if and only if $\mathbf{p} = \mathbf{q}$.*
2. *For all probability distributions $\mathbf{p}$ and $\mathbf{q}$ over $\mathcal{X}$, $\text{EMD}(\mathbf{p}, \mathbf{q}) = \text{EMD}(\mathbf{q}, \mathbf{p})$.*
3. *For all probability distributions $\mathbf{p}$, $\mathbf{q}$, and $\mathbf{r}$ over $\mathcal{X}$, $\text{EMD}(\mathbf{p}, \mathbf{r}) \leq \text{EMD}(\mathbf{p}, \mathbf{q}) + \text{EMD}(\mathbf{q}, \mathbf{r})$.*

There is another set of constraints on transports which yields an identical value of EMD but which can be more convenient to work with in certain cases. The conditions (2) and (2) imply the following weaker condition

$$(\mathbf{\Pi} - \mathbf{\Pi}^T)\mathbf{1} = \mathbf{p} - \mathbf{q}. \quad (6)$$

Denote by $\mathcal{D}(\mathbf{p}, \mathbf{q})$ the set of non-negative matrices $\mathbf{\Pi}$ satisfying (6). This condition only states that the change in mass at any location i is equal to $p_i - q_i$; it does not require the total mass leaving i to equal p_i , or the total mass entering i to equal q_i . It turns out, however, that minimising $\text{cost}(\mathbf{\Pi})$ over $\mathbf{\Pi}$ that satisfy this weaker condition yields EMD.

Theorem 1.2. *Let $\mathbf{p}$ and $\mathbf{q}$ be two probability vectors of length n , and let $\mathbf{D}$ be an n -by- n matrix of distances. Then*

$$\text{EMD}(\mathbf{p}, \mathbf{q}) = \min_{\mathbf{\Pi} \in \mathcal{D}(\mathbf{p}, \mathbf{q})} \text{cost}(\mathbf{\Pi}). \quad (7)$$

Another fundamental result on EMD is the *Kantorovich-Rubinstein Theorem*:

Theorem 1.3 (Kantorovich-Rubinstein). *Let $\mathbf{p}$ and $\mathbf{q}$ be two probability vectors of length n , and let $\mathbf{D}$ be an n -by- n matrix of distances. Let $\mathbf{\Lambda}$ denote the set of all vectors $\mathbf{x}$ satisfying $|x_i - x_j| \leq d_{ij}$ for all i and j . Then*

$$\text{EMD}(\mathbf{p}, \mathbf{q}) = \sup_{\mathbf{x} \in \mathbf{\Lambda}} \langle \mathbf{p} - \mathbf{q}, \mathbf{x} \rangle. \quad (8)$$

This result is a fairly straightforward corollary of duality in linear programming [1], one version of which states that (under very mild assumptions) the solution to the linear program

$$\text{minimize } \langle \mathbf{c}, \mathbf{x} \rangle \quad (9)$$

$$\text{s.t. } \mathbf{A}\mathbf{x} \leq \mathbf{b} \quad (10)$$

(where the inequality is understood elementwise) is equivalent to the *dual* linear program

$$\text{maximize } \langle \mathbf{b}, \mathbf{y} \rangle \quad (11)$$

$$\text{s.t. } \mathbf{A}^T \mathbf{y} - \mathbf{c} = \mathbf{0} \quad (12)$$

$$\mathbf{y} \leq \mathbf{0}. \quad (13)$$

1.2 Examples of EMD

1.2.1 The trivial metric

Suppose $\mathcal{X} = \{1, \dots, n\}$, and $d_{ij} = 0$ if $i = j$, and is 1 otherwise; that is, $\mathbf{D} = \mathbf{1}\mathbf{1}^T - \mathbf{I}$. Then $\text{EMD}(\mathbf{p}, \mathbf{q}) = \frac{1}{2} \|\mathbf{p} - \mathbf{q}\|_1$. We will prove this in two ways.

Proof #1. First, note that

$$\begin{aligned}
 \text{cost}(\mathbf{\Pi}) &= \langle \mathbf{1}\mathbf{1}^T, \mathbf{\Pi} \rangle - \langle \mathbf{I}, \mathbf{\Pi} \rangle \\
 &= \mathbf{1}^T \mathbf{\Pi} \mathbf{1} - \text{tr}(\mathbf{\Pi}) \\
 &= \mathbf{1}^T \mathbf{p} - \text{tr}(\mathbf{\Pi}) \\
 &= 1 - \text{tr}(\mathbf{\Pi}).
 \end{aligned} \tag{14}$$

Consequently, an optimal transport is one with maximal trace. For any index i , if $q_i \geq p_i$, then an optimal transport will not move mass out of i , and so in particular $\pi_{ii} = p_i$; and if $p_i \geq q_i$, an optimal transport will move exactly $p_i - q_i$ out of i and keep q_i mass at i , hence $\pi_{ii} = q_i$. Therefore, the cost of an optimal transport $\mathbf{\Pi}$ is

$$\begin{aligned}
 \text{cost}(\mathbf{\Pi}) &= 1 - \text{tr}(\mathbf{\Pi}) \\
 &= 1 - \sum_{i:p_i \leq q_i} p_i - \sum_{i:p_i \geq q_i} q_i \\
 &= \sum_{i:p_i \geq q_i} p_i - \sum_{i:p_i \geq q_i} q_i \\
 &= \sum_{i:p_i \geq q_i} (p_i - q_i).
 \end{aligned} \tag{15}$$

We conclude with the following lemma:

Lemma 1.4. *Let $\mathbf{p}$ and $\mathbf{q}$ be two probability vectors of length n . Then*

$$\|\mathbf{p} - \mathbf{q}\|_1 = 2 \sum_{i:p_i \geq q_i} (p_i - q_i). \tag{16}$$

Proof. Since $\mathbf{p}$ and $\mathbf{q}$ are both probability vectors,

$$\begin{aligned}
 0 &= \sum_{i=1}^n (p_i - q_i) \\
 &= \sum_{i:p_i \geq q_i} (p_i - q_i) + \sum_{i:p_i < q_i} (p_i - q_i), \\
 &= \sum_{i:p_i \geq q_i} (p_i - q_i) - \sum_{i:p_i < q_i} (q_i - p_i),
 \end{aligned} \tag{17}$$

and so

$$\sum_{i:p_i \geq q_i} (p_i - q_i) = \sum_{i:p_i < q_i} (q_i - p_i). \tag{18}$$

Consequently,

$$\begin{aligned}
 \|\mathbf{p} - \mathbf{q}\|_1 &= \sum_{i=1}^n |p_i - q_i| \\
 &= \sum_{i:p_i \geq q_i} (p_i - q_i) + \sum_{i:p_i < q_i} (q_i - p_i) \\
 &= \sum_{i:p_i \geq q_i} (p_i - q_i) + \sum_{i:p_i \geq q_i} (p_i - q_i) \\
 &= 2 \sum_{i:p_i \geq q_i} (p_i - q_i),
 \end{aligned} \tag{19}$$

as claimed. $\square$

Proof #2. Let Λ be the set of all vectors $\mathbf{x}$ with $|x_i - x_j| \leq d_{ij}$ for all i and j ; that is, $\mathbf{x}$ is in Λ if and only if $|x_i - x_j| \leq 1$ for all $i \neq j$. We start with a simple lemma characterizing Λ :

Lemma 1.5. *Let $\mathbf{x} \in \mathbb{R}^n$, and let*

$$c(\mathbf{x}) = \frac{\max_i x_i + \min_i x_i}{2}. \quad (20)$$

Then $|x_i - x_j| \leq 1$ for all $i \neq j$ if and only if $|x_i - c(\mathbf{x})| \leq 1/2$ for all i .

Proof. The implication $(\Leftarrow)$ is immediate from the triangle inequality. For the reverse, without loss of generality suppose $x_1 = \max_i x_i$ and $x_2 = \min_i x_i$, so that $c \equiv c(\mathbf{x}) = (x_1 + x_2)/2$. Then

$$|x_1 - c| = \frac{1}{2} |x_1 - x_2| \leq \frac{1}{2}, \quad (21)$$

and similarly $|x_2 - c| \leq 1/2$ too. But all other values $x_3, \dots, x_n$ either lie in the interval $[x_2, c]$ or $[c, x_1]$; hence, $|x_i - c| \leq 1/2$ for all i , as claimed. $\square$

Now, by Kantorovich-Rubinstein, the fact that any vector $\mathbf{x}$ can be shifted by a constant so that $c(\mathbf{x}) = 0$, and the fact that $\langle \mathbf{p} - \mathbf{q}, \mathbf{1} \rangle = 0$, we have:

$$\begin{aligned} \text{EMD}(\mathbf{p}, \mathbf{q}) &= \sup_{\mathbf{x} \in \Lambda} \langle \mathbf{p} - \mathbf{q}, \mathbf{x} \rangle \\ &= \sup_{\mathbf{x}: \|\mathbf{x}\|_\infty \leq 1/2, c(\mathbf{x})=0} \langle \mathbf{p} - \mathbf{q}, \mathbf{x} \rangle \\ &= \sup_{\mathbf{x}: \|\mathbf{x}\|_\infty \leq 1/2} \langle \mathbf{p} - \mathbf{q}, \mathbf{x} \rangle \\ &= \frac{1}{2} \sup_{\mathbf{x}: \|\mathbf{x}\|_\infty \leq 1} \langle \mathbf{p} - \mathbf{q}, \mathbf{x} \rangle \\ &= \frac{1}{2} \|\mathbf{p} - \mathbf{q}\|_1, \end{aligned} \quad (22)$$

which is the desired identity.

1.2.2 Distributions on the line

Let $t_1 < t_2 < \dots < t_n$, and let $d_{ij} = |t_i - t_j|$. Then

$$\text{EMD}(\mathbf{p}, \mathbf{q}) = \sum_{k=1}^n |t_{k+1} - t_k| \cdot \left| \sum_{j=1}^k (p_j - q_j) \right|. \quad (23)$$

In other words, $\text{EMD}(\mathbf{p}, \mathbf{q})$ is the L_1 distance between the cumulative distribution functions of $\mathbf{p}$ and $\mathbf{q}$.

Proof #1. First, we show that it is enough to consider transports that only move mass between adjacent locations.

Lemma 1.6. *Let $\Pi = (\pi_{ij})$ be a transport in $\mathcal{D}(\mathbf{p}, \mathbf{q})$. Suppose $1 \leq i < j \leq n$ with $j - i \geq 2$, and $\pi_{ij} > 0$. Let $\tilde{\Pi} = (\tilde{\pi}_{ij})$ be identical to Π , except $\tilde{\pi}_{ij} = 0$, $\tilde{\pi}_{i,i+1} = \pi_{ij} + \pi_{i,i+1}$ and $\tilde{\pi}_{i+1,j} = \pi_{ij} + \pi_{i+1,j}$. Then $\tilde{\Pi}$ is also in $\mathcal{D}(\mathbf{p}, \mathbf{q})$, and $\text{cost}(\tilde{\Pi}) = \text{cost}(\Pi)$.*

Proof. First, we show that $(\tilde{\Pi} - \Pi)^T \mathbf{1} = \mathbf{p} - \mathbf{q}$. The only rows and columns of $\tilde{\Pi}$ that are different from those of Π are the

i -th, $(i+1)$ -th, and j -th. For the i -th entry of $(\tilde{\mathbf{\Pi}} - \tilde{\mathbf{\Pi}}^T)\mathbf{1}$, we have:

$$\begin{aligned}
\sum_{k=1}^n (\tilde{\pi}_{ik} - \tilde{\pi}_{ki}) &= \sum_{k \neq i, i+1, j} (\tilde{\pi}_{ik} - \tilde{\pi}_{ki}) + (\tilde{\pi}_{i, i+1} - \tilde{\pi}_{i+1, i}) + (\tilde{\pi}_{ij} - \tilde{\pi}_{ji}) \\
&= \sum_{k \neq i, i+1, j} (\pi_{ik} - \pi_{ki}) + (\tilde{\pi}_{i, i+1} - \tilde{\pi}_{i+1, i}) + (\tilde{\pi}_{ij} - \tilde{\pi}_{ji}) \\
&= \sum_{k \neq i, i+1, j} (\pi_{ik} - \pi_{ki}) + (\pi_{ij} + \pi_{i, i+1} - \pi_{i+1, i}) + (0 - \pi_{ji}) \\
&= \sum_{k \neq i, i+1, j} (\pi_{ik} - \pi_{ki}) + (\pi_{ij} - \pi_{ji}) + (\pi_{i, i+1} - \pi_{i+1, i}) \\
&= \sum_{k=1}^n (\pi_{ik} - \pi_{ki}) \\
&= p_i - q_i.
\end{aligned} \tag{24}$$

For the $(i+1)$ -th entry of $(\tilde{\mathbf{\Pi}} - \tilde{\mathbf{\Pi}}^T)\mathbf{1}$, we have:

$$\begin{aligned}
\sum_{k=1}^n (\tilde{\pi}_{i+1, k} - \tilde{\pi}_{k, i+1}) &= \sum_{k \neq i, i+1, j} (\tilde{\pi}_{i+1, k} - \tilde{\pi}_{k, i+1}) + (\tilde{\pi}_{i+1, i} - \tilde{\pi}_{i, i+1}) + (\tilde{\pi}_{i+1, j} - \tilde{\pi}_{j, i+1}) \\
&= \sum_{k \neq i, i+1, j} (\pi_{i+1, k} - \pi_{k, i+1}) + (\pi_{i+1, i} - \tilde{\pi}_{i, i+1}) + (\tilde{\pi}_{i+1, j} - \pi_{j, i+1}) \\
&= \sum_{k \neq i, i+1, j} (\pi_{i+1, k} - \pi_{k, i+1}) + (\pi_{i+1, i} - [\pi_{ij} + \pi_{i, i+1}]) + (\pi_{ij} + \pi_{i+1, j} - \pi_{j, i+1}) \\
&= \sum_{k \neq i, i+1, j} (\pi_{i+1, k} - \pi_{k, i+1}) + (\pi_{i+1, i} - \pi_{i, i+1}) + (\pi_{i+1, j} - \pi_{j, i+1}) \\
&= \sum_{k=1}^n (\pi_{i+1, k} - \pi_{k, i+1}) \\
&= p_{i+1} - q_{i+1}.
\end{aligned} \tag{25}$$

Finally, for the j -th entry of $(\tilde{\mathbf{\Pi}} - \tilde{\mathbf{\Pi}}^T)\mathbf{1}$, we have:

$$\begin{aligned}
\sum_{k=1}^n (\tilde{\pi}_{i+1, k} - \tilde{\pi}_{k, i+1}) &= \sum_{k \neq i, i+1, j} (\tilde{\pi}_{jk} - \tilde{\pi}_{kj}) + (\tilde{\pi}_{ji} - \tilde{\pi}_{ij}) + (\tilde{\pi}_{j, i+1} - \tilde{\pi}_{i+1, j}) \\
&= \sum_{k \neq i, i+1, j} (\pi_{jk} - \pi_{kj}) + (\pi_{ji} - \tilde{\pi}_{ij}) + (\pi_{j, i+1} - \tilde{\pi}_{i+1, j}) \\
&= \sum_{k \neq i, i+1, j} (\pi_{jk} - \pi_{kj}) + (\pi_{ji} - 0) + (\pi_{j, i+1} - [\pi_{ij} + \pi_{i+1, j}]) \\
&= \sum_{k \neq i, i+1, j} (\pi_{jk} - \pi_{kj}) + (\pi_{ji} - \pi_{ij}) + (\pi_{j, i+1} - \pi_{i+1, j}) \\
&= \sum_{k=1}^n (\pi_{jk} - \pi_{kj}) \\
&= p_j - q_j.
\end{aligned} \tag{26}$$

This completes the argument that $\tilde{\mathbf{\Pi}}$ is in $\mathcal{D}(\mathbf{p}, \mathbf{q})$.

Next, we show that $\text{cost}(\tilde{\mathbf{\Pi}}) = \text{cost}(\mathbf{\Pi})$. We have:

$$\begin{aligned}
\tilde{\pi}_{i, i+1}(t_{i+1} - t_i) + \tilde{\pi}_{ij}(t_j - t_i) + \tilde{\pi}_{i+1, j}(t_j - t_{i+1}) &= (\pi_{ij} + \pi_{i, i+1})(t_{i+1} - t_i) + (\pi_{i+1, j} + \pi_{ij})(t_j - t_{i+1}) \\
&= \pi_{ij}(t_j - t_i) + \pi_{i, i+1}(t_{i+1} - t_i) + \pi_{i+1, j}(t_j - t_{i+1}),
\end{aligned} \tag{27}$$

and since all other entries of $\mathbf{\Pi}$ and $\tilde{\mathbf{\Pi}}$ are the same, this proves that $\text{cost}(\tilde{\mathbf{\Pi}}) = \text{cost}(\mathbf{\Pi})$, as desired. $\square$

By the lemma, it is enough to consider transports in $\mathcal{D}(\mathbf{p}, \mathbf{q})$ that move mass between adjacent locations only (that is, where $\mathbf{\Pi}$ is tridiagonal).

Lemma 1.7. *Let $\mathbf{\Pi}$ be the optimal tridiagonal transport in $\mathcal{D}(\mathbf{p}, \mathbf{q})$. Then for all $k = 1, \dots, n-1$,*

1. *If $p_1 + \dots + p_k \geq q_1 + \dots + q_k$, then $\pi_{k,k+1} = p_1 + \dots + p_k - (q_1 + \dots + q_k)$ and $\pi_{k+1,k} = 0$;*
2. *If $p_1 + \dots + p_k < q_1 + \dots + q_k$, then $\pi_{k+1,k} = q_1 + \dots + q_k - (p_1 + \dots + p_k)$ and $\pi_{k,k+1} = 0$.*

Proof. First, note that for the optimal transport, at most one of $\pi_{k,k+1}$ or $\pi_{k+1,k}$ is non-zero (sending mass from t_k to t_{k+1} and also from t_{k+1} to t_k is inefficient; fill in the details as an exercise).

Now, we will prove the result by induction on k . For $k = 1$, if $p_1 \geq q_1$, then $p_1 - q_1$ mass must be transferred from t_1 to t_2 . Consequently, $\pi_{12} = p_1 - q_1$, and $\pi_{21} = 0$. If $p_1 < q_1$, then $q_1 - p_1$ mass must be transferred from t_2 to t_1 , and so $\pi_{21} = q_1 - p_1$ and $\pi_{12} = 0$. This completes the proof of the base case.

Suppose the result is true for k ; we will show it for $k+1$. By the definition of $\mathcal{D}(\mathbf{p}, \mathbf{q})$ and the fact that $\mathbf{\Pi}$ is tridiagonal, we must have

$$\pi_{k,k-1} + \pi_{k,k+1} - (\pi_{k-1,k} + \pi_{k+1,k}) = p_k - q_k. \quad (28)$$

Suppose first that $\pi_{k,k-1} > 0$. By the inductive hypothesis, this means $\pi_{k-1,k} = 0$, and $\pi_{k,k-1} = q_1 + \dots + q_{k-1} - (p_1 + \dots + p_{k-1})$. Consequently, (28) becomes:

$$\pi_{k,k+1} - \pi_{k+1,k} = p_1 + \dots + p_k - (q_1 + \dots + q_k). \quad (29)$$

Now, if $p_1 + \dots + p_k \geq q_1 + \dots + q_k$, then because $\pi_{k,k+1} \geq 0$ and $\pi_{k+1,k} \geq 0$, and at most one can be non-positive, we must have

$$\pi_{k,k+1} = p_1 + \dots + p_k - (q_1 + \dots + q_k). \quad (30)$$

On the other hand, if $p_1 + \dots + p_k < q_1 + \dots + q_k$, then the same reasoning shows

$$\pi_{k+1,k} = q_1 + \dots + q_k - (p_1 + \dots + p_k). \quad (31)$$

A nearly identical proof applies if $\pi_{k-1,k} > 0$. This completes the proof $\square$

Since the cost of moving mass between t_k and t_{k+1} is $t_{k+1} - t_k$, and because at most one of $\pi_{k,k+1}$ and $\pi_{k+1,k}$ is non-zero, the preceding lemma immediately yields the desired formula.

Proof #2. Recall the summation-by-parts formula:

Lemma 1.8. *Suppose $\mathbf{x} = (x_0, \dots, x_n)$ and $\mathbf{y} = (y_0, \dots, y_n)$ are two vectors of length $n+1$. Then*

$$\sum_{i=1}^n x_i(y_i - y_{i-1}) = x_n y_n - x_0 y_0 - \sum_{i=1}^n (x_i - x_{i-1}) y_{i-1}. \quad (32)$$

Proof. We have:

$$\begin{aligned} \sum_{i=1}^n x_i(y_i - y_{i-1}) &= \sum_{i=1}^n (x_i - x_{i-1})(y_i - y_{i-1}) + \sum_{i=1}^n x_{i-1}(y_i - y_{i-1}) \\ &= -\sum_{i=1}^n (x_i - x_{i-1})y_{i-1} + \sum_{i=1}^n (x_i - x_{i-1})y_i + \sum_{i=1}^n x_{i-1}(y_i - y_{i-1}) \\ &= -\sum_{i=1}^n (x_i - x_{i-1})y_{i-1} + \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_{i-1} y_{i-1} \\ &= -\sum_{i=1}^n (x_i - x_{i-1})y_{i-1} + x_n y_n - x_0 y_0, \end{aligned} \quad (33)$$

as claimed. $\square$

We will also use the following lemma characterizing Lipschitz vectors:

Lemma 1.9. *Let $\mathbf{x} \in \mathbb{R}^n$. Then $\mathbf{x} \in \mathbf{\Lambda}$ if and only if $|x_{i+1} - x_i| \leq d_{i+1,i}$ for all $i = 1, \dots, n-1$.*

Proof. The implication $(\Rightarrow)$ is immediate. For the reverse, take any $i < j$, so that $t_i < t_j$. Then

$$\begin{aligned} |x_j - x_i| &\leq |x_i - x_{i+1}| + |x_{i+1} - x_{i+2}| + \dots + |x_{j-1} - x_j| \\ &\leq d_{i,i+1} + d_{i+1,i+2} + \dots + d_{j-1,j} \\ &= (t_{i+1} - t_i) + (t_{i+2} - t_{i+1}) + \dots + (t_j - t_{j-1}) \\ &= t_j - t_i \\ &= d_{ij}, \end{aligned} \tag{34}$$

which is the desired result. $\square$

We now prove formula (23) for $\text{EMD}(\mathbf{p}, \mathbf{q})$. Let $\boldsymbol{\delta} = \mathbf{p} - \mathbf{q}$. Define the vector $\mathbf{s} = (s_0, \dots, s_n)$ with entries $s_0 = 0$ and

$$s_k = \sum_{j=1}^k \delta_j, \quad 1 \leq k \leq n. \tag{35}$$

Note that $s_0 = s_n = 0$, and that $\delta_k = s_k - s_{k-1}$ for $1 \leq k \leq n$. Consequently, for any vector $\mathbf{x}$ in $\mathbb{R}^{n+1}$, we have

$$\begin{aligned} \langle \boldsymbol{\delta}, \mathbf{x} \rangle &= \sum_{k=1}^n \delta_k x_k \\ &= \sum_{k=1}^n (s_k - s_{k-1}) x_k \\ &= s_n x_n - s_0 x_0 - \sum_{k=1}^n (x_k - x_{k-1}) s_{k-1} \\ &= - \sum_{k=1}^n (x_k - x_{k-1}) s_{k-1} \\ &= - \sum_{k=2}^n (x_k - x_{k-1}) s_{k-1}. \end{aligned} \tag{36}$$

By Lemma 1.9, $\mathbf{x} \in \mathbf{\Lambda}$ if and only if the vector

$$\boldsymbol{\Delta}_{\mathbf{x}} = \left(\frac{x_2 - x_1}{t_2 - t_1}, \frac{x_3 - x_2}{t_3 - t_2}, \dots, \frac{x_n - x_{n-1}}{t_n - t_{n-2}} \right) \tag{37}$$

has ℓ_∞ norm less than or equal to 1. Consequently, by the Kantorovich-Rubinstein Theorem, and noting again that $s_n = s_0 = 1$,

$$\begin{aligned} \text{EMD}(\mathbf{p}, \mathbf{q}) &= \sup_{\mathbf{x} \in \mathbf{\Lambda}} \langle \boldsymbol{\delta}, \mathbf{x} \rangle \\ &= \sup_{\mathbf{x} \in \mathbf{\Lambda}} \sum_{k=2}^n (x_k - x_{k-1}) s_{k-1} \\ &= \sup_{\mathbf{x} \in \mathbf{\Lambda}} \sum_{k=2}^n \frac{x_k - x_{k-1}}{t_k - t_{k-1}} s_{k-1} (t_k - t_{k-1}) \\ &= \sup_{\boldsymbol{\Delta}: \|\boldsymbol{\Delta}\|_\infty \leq 1} \sum_{k=2}^n \Delta_{k-1} s_{k-1} (t_k - t_{k-1}) \\ &= \sum_{k=1}^n |s_{k-1}| |t_k - t_{k-1}|, \end{aligned} \tag{38}$$

which is the desired result.

2 Sinkhorn's algorithm

Suppose $\mathbf{A} = (a_{ij})$ is a positive matrix of size m -by- n , and $\mathbf{r} = (r_1, \dots, r_m)$ and $\mathbf{c} = (c_1, \dots, c_n)$ are vectors of lengths m and n , respectively, with positive entries: $r_i > 0$ and $c_j > 0$. Suppose too that $\mathbf{r}$ and $\mathbf{c}$ are normalized so that

$$\sum_{i=1}^m r_i = \sum_{j=1}^n c_j. \quad (39)$$

We ask the following question: is it possible to find diagonal matrices $\mathbf{D}_1$ and $\mathbf{D}_2$ such that the row sums of $\mathbf{D}_1 \mathbf{A} \mathbf{D}_2$ are equal to $r_1, \dots, r_m$, and the columns sums of $\mathbf{D}_1 \mathbf{A} \mathbf{D}_2$ are equal to $c_1, \dots, c_n$? It turns out that the answer is yes, and furthermore, there is a straightforward iterative algorithm for computing $\mathbf{D}_1$ and $\mathbf{D}_2$. We first give the algorithm, and will then prove that it produces the desired solution. The algorithm and proof may be found in [5].

1. Let $\tilde{\mathbf{x}}^{(0)} = \mathbf{1}$ and $\tilde{\mathbf{y}}^{(0)} = (\tilde{y}_1^{(0)}, \dots, \tilde{y}_n^{(0)})$ be defined by

$$\tilde{y}_j^{(0)} = \frac{c_j}{\sum_{i=1}^m a_{ij}}. \quad (40)$$

2. For $k = 0, 1, 2, \dots$, define $\tilde{\mathbf{x}}^{(k+1)} = (\tilde{x}_1^{(k+1)}, \dots, \tilde{x}_m^{(k+1)})$ by

$$\tilde{x}_i^{(k+1)} = \frac{r_i}{\sum_{j=1}^n a_{ij} \tilde{y}_j^{(k)}}, \quad (41)$$

and define $\tilde{\mathbf{y}}^{(k+1)} = (\tilde{y}_1^{(k+1)}, \dots, \tilde{y}_n^{(k+1)})$ by

$$\tilde{y}_j^{(k+1)} = \frac{c_j}{\sum_{i=1}^m \tilde{x}_i^{(k+1)} a_{ij}}. \quad (42)$$

Theorem 2.1 (Sinkhorn). *As $k \rightarrow \infty$, the sequence of matrices $\text{diag}(\tilde{\mathbf{x}}^{(k)}) \mathbf{A} \text{diag}(\tilde{\mathbf{y}}^{(k)})$ converges to a matrix of the form $\text{diag}(\mathbf{x}) \mathbf{A} \text{diag}(\mathbf{y})$ satisfying*

$$\sum_{j=1}^n x_i a_{ij} y_j = r_i, \quad 1 \leq i \leq m, \quad (43)$$

and

$$\sum_{i=1}^m x_i a_{ij} y_j = c_j, \quad 1 \leq j \leq n. \quad (44)$$

Furthermore, the limiting vectors $\mathbf{x}$ and $\mathbf{y}$ are unique up to rescaling.

First we prove uniqueness. Suppose $\text{diag}(\mathbf{x}) \mathbf{A} \text{diag}(\mathbf{y})$ and $\text{diag}(\tilde{\mathbf{x}}) \mathbf{A} \text{diag}(\tilde{\mathbf{y}})$ both have row and column sums $\mathbf{r}$ and $\mathbf{c}$, respectively. Let

$$p_i = \frac{\tilde{x}_i}{x_i}, \quad 1 \leq i \leq m, \quad (45)$$

and

$$q_j = \frac{\tilde{y}_j}{y_j}, \quad 1 \leq j \leq n. \quad (46)$$

It is enough to show that $p_i = p_{i'}$ for all $1 \leq i, i' \leq m$, and $q_j = q_{j'}$ for all $1 \leq j, j' \leq n$. Suppose $p_{i_1} = \min_i p_i$ and $q_{j_1} = \max_j q_j$. Then

$$r_i = \sum_{j=1}^n \tilde{x}_i a_{ij} \tilde{y}_j, \quad (47)$$

and so

$$\tilde{x}_i = \frac{r_i}{\sum_{j=1}^n a_{ij} \tilde{y}_j}. \quad (48)$$

Applying this to $i = i_1$ gives

$$\begin{aligned} p_{i_1} &= \frac{\tilde{x}_{i_1}}{x_{i_1}} \\ &= \frac{r_{i_1}}{\sum_{j=1}^n x_{i_1} a_{i_1 j} \tilde{y}_j} \\ &= \frac{r_{i_1}}{\sum_{j=1}^n x_{i_1} a_{i_1 j} y_j q_j} \\ &\geq \frac{1}{q_{j_1}} \frac{r_{i_1}}{\sum_{j=1}^n x_{i_1} a_{i_1 j} y_j} \\ &= \frac{1}{q_{j_1}}, \end{aligned} \quad (49)$$

and hence

$$\frac{1}{p_{i_1}} \leq q_{j_1}. \quad (50)$$

On the other hand, we also have

$$\tilde{y}_j = \frac{c_j}{\sum_{i=1}^m \tilde{x}_i a_{ij}}, \quad (51)$$

and therefore

$$\begin{aligned} q_{j_1} &= \frac{\tilde{y}_{j_1}}{y_{j_1}} \\ &= \frac{c_{j_1}}{\sum_{i=1}^m \tilde{x}_i a_{ij_1} y_{j_1}} \\ &= \frac{c_{j_1}}{\sum_{i=1}^m p_i x_i a_{ij_1} y_{j_1}} \\ &\leq \frac{1}{p_{i_1}} \frac{c_{j_1}}{\sum_{i=1}^m x_i a_{ij_1} y_{j_1}} \\ &= \frac{1}{p_{i_1}} \\ &\leq q_{j_1} \end{aligned} \quad (52)$$

where we have used (50) for the last inequality. Therefore, the inequalities in (49) and (52) are equalities, which implies that $p_i = p_{i_1}$ for all $1 \leq i \leq m$ and $q_j = q_{j_1}$ for all $1 \leq j \leq n$; hence $p_i = p_{i'}$ for all $1 \leq i, i' \leq m$, and $q_j = q_{j'}$ for all $1 \leq j, j' \leq n$; this proves uniqueness.

We now show that the matrices $\text{diag}(\tilde{\mathbf{x}}^{(k)}) \mathbf{A} \text{diag}(\tilde{\mathbf{y}}^{(k)})$ converge to a matrix $\text{diag}(\mathbf{x}) \mathbf{A} \text{diag}(\mathbf{y})$ with the specified row and column sums. It is easier to work with rescaled vectors $\mathbf{x}^{(k)}$ and $\mathbf{y}^{(k)}$, defined by

$$\mathbf{x}^{(k)} = \frac{1}{\|\tilde{\mathbf{x}}^{(k)}\|_\infty} \tilde{\mathbf{x}}^{(k)}, \quad (53)$$

and

$$\mathbf{y}^{(k)} = \|\tilde{\mathbf{x}}^{(k)}\|_\infty \tilde{\mathbf{y}}^{(k)}. \quad (54)$$

Without loss of generality, we will assume that $\mathbf{A}$, $\mathbf{r}$ and $\mathbf{c}$ are scaled so that

$$1 = \min_{i,j} \{a_{ij}\} = \max_{i,j} \{r_i, c_j\}. \quad (55)$$

Note that $\text{diag}(\tilde{\mathbf{x}}^{(k)}) \mathbf{A} \text{diag}(\tilde{\mathbf{y}}^{(k)}) = \text{diag}(\mathbf{x}^{(k)}) \mathbf{A} \text{diag}(\mathbf{y}^{(k)})$, and so in particular $\text{diag}(\mathbf{x}^{(k)}) \mathbf{A} \text{diag}(\mathbf{y}^{(k)})$ has column sums $c_1, \dots, c_n$. Furthermore, with the vectors scaled in this way, we have the following lemma:

Lemma 2.2. For all k , $\|\mathbf{x}^{(k)}\|_\infty = 1$ and $\|\mathbf{y}^{(k)}\|_\infty \leq 1$.

Proof. The identity $\|\mathbf{x}^{(k)}\|_\infty = 1$ is immediate. For any $1 \leq j \leq n$, we have

$$\sum_{i=1}^m x_i^{(k)} a_{ij} y_j^{(k)} = c_j. \quad (56)$$

Consequently, for all $1 \leq i \leq m$ and $1 \leq j \leq n$, we have

$$\begin{aligned} y_j^{(k)} &= \frac{c_j}{\sum_{i'=1}^m x_{i'}^{(k)} a_{i'j}} \\ &\leq \frac{c_j}{x_i^{(k)} a_{ij}} \\ &\leq \frac{1}{x_i^{(k)}}. \end{aligned} \quad (57)$$

Since this is true for all i , we can take i so that $x_i^{(k)} = \|\mathbf{x}^{(k)}\|_\infty = 1$, which completes the proof. $\square$

Because for each k the matrix $\text{diag}(\mathbf{x}^{(k)})\mathbf{A}\text{diag}(\mathbf{y}^{(k)})$ has column sums $c_1, \dots, c_n$, we must show that the row sums converge to $r_1, \dots, r_m$. To that end, define the function

$$\phi(\mathbf{x}, \mathbf{y}) = \max_{1 \leq i \leq m} \frac{\sum_{j=1}^n x_i a_{ij} y_j}{r_i} - \min_{1 \leq i \leq m} \frac{\sum_{j=1}^n x_i a_{ij} y_j}{r_i}. \quad (58)$$

It is convenient to define the following values, for each $k \geq 0$:

$$\alpha_i^{(k)} = \frac{\sum_{j=1}^n x_i^{(k)} a_{ij} y_j^{(k)}}{r_i}, \quad 1 \leq i \leq m. \quad (59)$$

Note that, by definition,

$$\phi(\mathbf{x}^{(k)}, \mathbf{y}^{(k)}) = \max_{1 \leq i \leq m} \alpha_i^{(k)} - \min_{1 \leq i \leq m} \alpha_i^{(k)}. \quad (60)$$

For $k \geq 0$, we also define the values

$$\beta_j^{(k)} = \frac{\sum_{i=1}^m x_i^{(k)} a_{ij} y_j^{(k)} / \alpha_i^{(k)}}{c_j}, \quad 1 \leq j \leq n, \quad (61)$$

and the value

$$M_k = \max_{1 \leq i \leq m} \frac{x_i^{(k)}}{\alpha_i^{(k)}}. \quad (62)$$

We then have the following lemma:

Lemma 2.3. For each $k \geq 0$, $1 \leq i \leq m$, and $1 \leq j \leq n$,

$$x_i^{(k+1)} = \frac{1}{M_k} \frac{x_i^{(k)}}{\alpha_i^{(k)}} \quad (63)$$

and

$$y_j^{(k+1)} = M_k \frac{y_j^{(k)}}{\beta_j^{(k)}}. \quad (64)$$

Proof. First, observe that

$$\begin{aligned}
\frac{x_i^{(k)}}{\alpha_i^{(k)}} &= x_i^{(k)} \frac{r_i}{\sum_{j=1}^n x_i^{(k)} a_{ij} y_j^{(k)}} \\
&= \frac{r_i}{\sum_{j=1}^n a_{ij} y_j^{(k)}} \\
&= \frac{1}{\|\tilde{\mathbf{x}}^{(k)}\|_\infty} \frac{r_i}{\sum_{j=1}^n a_{ij} \tilde{y}_j^{(k)}} \\
&= \frac{1}{\|\tilde{\mathbf{x}}^{(k)}\|_\infty} \tilde{x}_i^{(k+1)} \\
&= \frac{\|\tilde{\mathbf{x}}^{(k+1)}\|_\infty}{\|\tilde{\mathbf{x}}^{(k)}\|_\infty} x_i^{(k+1)}.
\end{aligned} \tag{65}$$

That is, if $C_k = \|\tilde{\mathbf{x}}^{(k)}\|_\infty^{-1} \|\tilde{\mathbf{x}}^{(k+1)}\|_\infty$, then

$$x_i^{(k+1)} = \frac{1}{C_k} \frac{x_i^{(k)}}{\alpha_i^{(k)}}. \tag{66}$$

Because $\|\mathbf{x}^{(k+1)}\|_\infty = 1$ and

$$\max_{1 \leq i \leq m} \frac{1}{M_k} \frac{x_i^{(k)}}{\alpha_i^{(k)}} = 1 \tag{67}$$

as well, it must be that $C_k = M_k$, which proves (63).

To prove (64), observe that

$$\begin{aligned}
\frac{y_j^{(k)}}{\beta_j^{(k)}} &= \frac{y_j^{(k)} c_j}{\sum_{i=1}^m x_i^{(k)} a_{ij} y_j^{(k)} / \alpha_i^{(k)}} \\
&= \frac{c_j}{\sum_{i=1}^m x_i^{(k)} a_{ij} / \alpha_i^{(k)}} \\
&= \frac{1}{M_k} \frac{c_j}{\sum_{i=1}^m x_i^{(k+1)} a_{ij}} \\
&= \frac{1}{M_k} y_j^{(k+1)},
\end{aligned} \tag{68}$$

which yields the desired identity. □

Corollary 2.4. For each $k \geq 0$, $1 \leq i \leq m$, and $1 \leq j \leq n$,

$$\alpha_i^{(k)} \beta_j^{(k)} = \frac{x_i^{(k)} y_j^{(k)}}{x_i^{(k+1)} y_j^{(k+1)}}. \tag{69}$$

Proof. This is immediate from Lemma 2.3. □

Lemma 2.5. For all $k \geq 0$ and $1 \leq j \leq n$,

$$\min_{1 \leq i \leq m} \frac{1}{\alpha_i^{(k)}} \leq \beta_j^{(k)} \leq \max_{1 \leq i \leq m} \frac{1}{\alpha_i^{(k)}}. \tag{70}$$

Proof. From Lemma 2.3 we have

$$x_i^{(k+1)} = \frac{1}{M_k} \frac{x_i^{(k)}}{\alpha_i^{(k)}} \tag{71}$$

and

$$y_j^{(k+1)} = M_k \frac{y_j^{(k)}}{\beta_j^{(k)}}. \quad (72)$$

Consequently, because $\sum_{i=1}^m x_i^{(k+1)} a_{ij} y_j^{(k+1)} = c_j = \sum_{i=1}^m x_i^{(k)} a_{ij} y_j^{(k)}$,

$$\begin{aligned} \beta_j^{(k)} &= M_k \frac{y_j^{(k)}}{y_j^{(k+1)}} \\ &= M_k \frac{\sum_{i=1}^m x_i^{(k+1)} a_{ij}}{\sum_{i=1}^m x_i^{(k)} a_{ij}} \\ &= \frac{\sum_{i=1}^m x_i^{(k)} a_{ij} / \alpha_i^{(k)}}{\sum_{i=1}^m x_i^{(k)} a_{ij}}, \end{aligned} \quad (73)$$

which is bounded below by $\min_i 1/\alpha_i^{(k)}$ and bounded above by $\max_i 1/\alpha_i^{(k)}$. $\square$

We now proceed to the main argument. Fix a value $k \geq 0$, and choose indices i_1 , i_2 , j_1 , and j_2 so that

$$\alpha_{i_1}^{(k+1)} = \min_{1 \leq i \leq m} \alpha_i^{(k+1)}, \quad (74)$$

$$\alpha_{i_2}^{(k+1)} = \max_{1 \leq i \leq m} \alpha_i^{(k+1)}, \quad (75)$$

$$\beta_{j_1}^{(k)} = \min_{1 \leq j \leq n} \beta_j^{(k)}, \quad (76)$$

and

$$\beta_{j_2}^{(k)} = \max_{1 \leq j \leq n} \beta_j^{(k)}. \quad (77)$$

Using Corollary 2.4, we have:

$$\begin{aligned} \phi(\mathbf{x}^{(k+1)}, \mathbf{y}^{(k+1)}) &= \alpha_{i_2}^{(k+1)} - \alpha_{i_1}^{(k+1)} \\ &= \sum_{j=1}^n \frac{x_{i_2}^{(k+1)} a_{i_2 j} y_j^{(k+1)}}{r_{i_2}} - \sum_{j=1}^n \frac{x_{i_1}^{(k+1)} a_{i_1 j} y_j^{(k+1)}}{r_{i_1}} \\ &= \sum_{j=1}^n \frac{x_{i_2}^{(k)} a_{i_2 j} y_j^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)} \beta_j^{(k)}} - \sum_{j=1}^n \frac{x_{i_1}^{(k)} a_{i_1 j} y_j^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)} \beta_j^{(k)}} \\ &= \sum_{j \neq j_2} \frac{x_{i_2}^{(k)} a_{i_2 j} y_j^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)} \beta_j^{(k)}} - \sum_{j \neq j_1} \frac{x_{i_1}^{(k)} a_{i_1 j} y_j^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)} \beta_j^{(k)}} + \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)} \beta_{j_2}^{(k)}} - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)} \beta_{j_1}^{(k)}} \\ &\leq \sum_{j \neq j_2} \frac{x_{i_2}^{(k)} a_{i_2 j} y_j^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)} \beta_{j_1}^{(k)}} - \sum_{j \neq j_1} \frac{x_{i_1}^{(k)} a_{i_1 j} y_j^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)} \beta_{j_2}^{(k)}} + \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)} \beta_{j_2}^{(k)}} - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)} \beta_{j_1}^{(k)}}. \end{aligned} \quad (78)$$

Now, observe that

$$\alpha_{i_2}^{(k)} = \frac{\sum_{j=1}^n x_{i_2}^{(k)} a_{i_2 j} y_j^{(k)}}{r_{i_2}} \quad (79)$$

and hence

$$\begin{aligned} 1 &= \sum_{j=1}^n \frac{x_{i_2}^{(k)} a_{i_2 j} y_j^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)}} \\ &= \sum_{j \neq j_2} \frac{x_{i_2}^{(k)} a_{i_2 j} y_j^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)}} + \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)}} \end{aligned} \quad (80)$$

from which we immediately find

$$\sum_{j \neq j_2} \frac{x_{i_2}^{(k)} a_{i_2 j} y_j^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)} \beta_{j_1}^{(k)}} = \frac{1}{\beta_{j_1}^{(k)}} \left(1 - \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)}} \right). \quad (81)$$

Similarly,

$$\sum_{j \neq j_1} \frac{x_{i_1}^{(k)} a_{i_1 j} y_j^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)} \beta_{j_2}^{(k)}} = \frac{1}{\beta_{j_2}^{(k)}} \left(1 - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)}} \right). \quad (82)$$

Using these identities and continuing (78), we get:

$$\begin{aligned} \phi(\mathbf{x}^{(k+1)}, \mathbf{y}^{(k+1)}) &\leq \sum_{j \neq j_2} \frac{x_{i_2}^{(k)} a_{i_2 j} y_j^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)} \beta_{j_1}^{(k)}} - \sum_{j \neq j_1} \frac{x_{i_1}^{(k)} a_{i_1 j} y_j^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)} \beta_{j_2}^{(k)}} + \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)} \beta_{j_2}^{(k)}} - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)} \beta_{j_1}^{(k)}} \\ &= \frac{1}{\beta_{j_1}^{(k)}} \left(1 - \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)}} \right) - \frac{1}{\beta_{j_2}^{(k)}} \left(1 - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)}} \right) + \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)} \beta_{j_2}^{(k)}} - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)} \beta_{j_1}^{(k)}} \\ &= \left(\frac{1}{\beta_{j_1}^{(k)}} - \frac{1}{\beta_{j_2}^{(k)}} \right) \left(1 - \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)}} - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)}} \right). \end{aligned} \quad (83)$$

Applying Lemma 2.5 we can bound the factor in parentheses on the left:

$$\begin{aligned} \phi(\mathbf{x}^{(k+1)}, \mathbf{y}^{(k+1)}) &\leq \left(\frac{1}{\beta_{j_1}^{(k)}} - \frac{1}{\beta_{j_2}^{(k)}} \right) \left(1 - \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)}} - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)}} \right) \\ &\leq \left(\max_{1 \leq i \leq n} \alpha_i^{(k)} - \min_{1 \leq i \leq n} \alpha_i^{(k)} \right) \left(1 - \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)}} - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)}} \right) \\ &= \phi(\mathbf{x}^{(k)}, \mathbf{y}^{(k)}) \left(1 - \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)}} - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)}} \right). \end{aligned} \quad (84)$$

We will now bound the factor in parentheses on the right. Define

$$\rho = \min\{c_1, \dots, c_n\}, \quad (85)$$

and

$$R = \sum_{i=1}^m \sum_{j=1}^n a_{ij}. \quad (86)$$

Then for all $1 \leq i \leq m$, by definition of $\alpha_i^{(k)}$ and the fact that $\|\mathbf{y}^{(k)}\|_\infty \leq 1$, we have:

$$\begin{aligned} \frac{x_i^{(k)}}{\alpha_i^{(k)}} &= \frac{r_i}{\sum_{j=1}^n a_{ij} y_j^{(k)}} \\ &\geq \frac{r_i}{\sum_{j=1}^n a_{ij}} \\ &= \frac{r_i}{R}. \end{aligned} \quad (87)$$

Consequently, we have the following lower bound:

$$\frac{x_i^{(k)}}{r_i \alpha_i^{(k)}} \geq \frac{1}{R}. \quad (88)$$

Furthermore, for all $1 \leq j \leq n$, using that $\|\mathbf{x}^{(k)}\|_\infty = 1$ we also have the lower bound

$$\begin{aligned} y_j^{(k)} &= \frac{c_j}{\sum_{i=1}^m x_i^{(k)} a_{ij}} \\ &\geq \frac{c_j}{\sum_{i=1}^m a_{ij}} \\ &\geq \frac{\rho}{R}. \end{aligned} \quad (89)$$

Putting (88) and (89) together and using that $a_{ij} \geq 1$ for all $1 \leq i \leq m$ and $1 \leq j \leq n$ gives

$$\frac{x_i^{(k)} a_{ij} y_j^{(k)}}{r_i \alpha_i^{(k)}} \geq \frac{\rho}{R^2}. \quad (90)$$

Using this bound, we continue (84):

$$\begin{aligned} \phi(\mathbf{x}^{(k+1)}, \mathbf{y}^{(k+1)}) &\leq \phi(\mathbf{x}^{(k)}, \mathbf{y}^{(k)}) \left(1 - \frac{x_{i_2}^{(k)} a_{i_2 j_2} y_{j_2}^{(k)}}{r_{i_2} \alpha_{i_2}^{(k)}} - \frac{x_{i_1}^{(k)} a_{i_1 j_1} y_{j_1}^{(k)}}{r_{i_1} \alpha_{i_1}^{(k)}} \right) \\ &\leq \phi(\mathbf{x}^{(k)}, \mathbf{y}^{(k)}) \left(1 - 2 \frac{\rho}{R^2} \right). \end{aligned} \quad (91)$$

Consequently, $\phi(\mathbf{x}^{(k)}, \mathbf{y}^{(k)}) \rightarrow 0$ as $k \rightarrow \infty$.

Because $\|(\mathbf{x}^{(k)}, \mathbf{y}^{(k)})\|_\infty = 1$ for all k , there is a subsequence $\{(\mathbf{x}^{(k_\ell)}, \mathbf{y}^{(k_\ell)})\}_\ell$ that converges to some $(\mathbf{x}, \mathbf{y})$. By the continuity of ϕ ,

$$\phi(\mathbf{x}, \mathbf{y}) = \lim_{\ell \rightarrow \infty} \phi(\mathbf{x}^{(k_\ell)}, \mathbf{y}^{(k_\ell)}) = 0. \quad (92)$$

From the uniqueness of solutions to the rescaling problem, it follows that, in fact, the entire sequence $(\mathbf{x}^{(k)}, \mathbf{y}^{(k)})$ converges to $(\mathbf{x}, \mathbf{y})$. Indeed, if not then there would be a subsequence whose distance to $(\mathbf{x}, \mathbf{y})$ is bounded from zero, and then a subsequence $\{(\mathbf{x}^{(m_\ell)}, \mathbf{y}^{(m_\ell)})\}_\ell$ of this subsequence must converge to some $(\mathbf{x}', \mathbf{y}') \neq (\mathbf{x}, \mathbf{y})$. But then $\phi(\mathbf{x}^{(m_\ell)}, \mathbf{y}^{(m_\ell)}) \rightarrow \phi(\mathbf{x}', \mathbf{y}')$ as $\ell \rightarrow \infty$, and so $\phi(\mathbf{x}', \mathbf{y}') = 0$ as well, hence $\text{diag}(\mathbf{x}') \mathbf{A} \text{diag}(\mathbf{y}') = \text{diag}(\mathbf{x}) \mathbf{A} \text{diag}(\mathbf{y})$, which means $\mathbf{x} = c \mathbf{x}'$ and $\mathbf{y} = c^{-1} \mathbf{y}'$ for some constant c . However, by continuity of the infinity norm, $\|\mathbf{x}'\|_\infty = 1 = \|\mathbf{x}\|_\infty$ (since $\|\mathbf{x}^{(k)}\|_\infty = 1$ for all k). Therefore, $c = 1$ and $(\mathbf{x}, \mathbf{y}) = (\mathbf{x}', \mathbf{y}')$, a contradiction.

3 Entropy-regularized EMD

The Earth Mover's Distance $\text{EMD}(\mathbf{p}, \mathbf{q})$ is defined as

$$\begin{aligned} \text{EMD}(\mathbf{p}, \mathbf{q}) &= \min_{\mathbf{\Pi} \in \mathcal{P}(\mathbf{p}, \mathbf{q})} \text{cost}(\mathbf{\Pi}) \\ &= \min_{\mathbf{\Pi} \in \mathcal{P}(\mathbf{p}, \mathbf{q})} \langle \mathbf{\Pi}, \mathbf{D} \rangle, \end{aligned} \quad (93)$$

where $\mathbf{D}$ is the matrix of distances. The idea behind the entropy-regularized approximation to EMD is to modify the objective to penalize entries that are too small. The *entropy* of a positive matrix $\mathbf{\Pi}$ is defined as

$$h(\mathbf{\Pi}) = \sum_{i,j} \pi_{ij} \log(1/\pi_{ij}). \quad (94)$$

The entropy-regularized EMD, or *Sinkhorn distance*, is then defined as

$$d_\lambda(\mathbf{p}, \mathbf{q}) = \min_{\mathbf{\Pi} \in \mathcal{P}(\mathbf{p}, \mathbf{q})} \langle \mathbf{\Pi}, \mathbf{D} \rangle - \frac{1}{\lambda} h(\mathbf{\Pi}), \quad (95)$$

where $\lambda > 0$. When λ is small, $d_\lambda(\mathbf{p}, \mathbf{q})$ will be close to $\text{EMD}(\mathbf{p}, \mathbf{q})$. However, as we will see, $d_\lambda(\mathbf{p}, \mathbf{q})$ can be computed efficiently using Sinkhorn's algorithm. Indeed, letting $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ be vectors of Lagrange multipliers, the Lagrangian of the problem (95) is the function

$$\begin{aligned}\mathcal{L}(\boldsymbol{\Pi}, \boldsymbol{\alpha}, \boldsymbol{\beta}) &= \langle \boldsymbol{\Pi}, \mathbf{D} \rangle - \frac{1}{\lambda} h(\boldsymbol{\Pi}) + \boldsymbol{\alpha}^T (\boldsymbol{\Pi} \mathbf{1} - \mathbf{p}) + \boldsymbol{\beta}^T (\boldsymbol{\Pi}^T \mathbf{1} - \mathbf{q}) \\ &= \sum_{i,j} \left\{ \pi_{ij} d_{ij} + \frac{1}{\lambda} \pi_{ij} \log(\pi_{ij}) + \alpha_i \pi_{ij} + \beta_j \pi_{ij} \right\} - \boldsymbol{\alpha}^T \mathbf{p} - \boldsymbol{\beta}^T \mathbf{q}.\end{aligned}\quad (96)$$

The partial derivative of $\mathcal{L}$ with respect to π_{ij} is then

$$\frac{\partial \mathcal{L}}{\partial \pi_{ij}}(\boldsymbol{\Pi}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = d_{ij} + \frac{1}{\lambda} + \frac{1}{\lambda} \log(\pi_{ij}) + \alpha_i + \beta_j. \quad (97)$$

Setting this equal to 0 and multiplying by λ gives the equation

$$0 = \lambda d_{ij} + 1 + \log(\pi_{ij}) + \lambda \alpha_i + \lambda \beta_j, \quad (98)$$

and exponentiating each side then gives

$$1 = \pi_{ij} \exp\{\lambda d_{ij} + 1 + \lambda \alpha_i + \lambda \beta_j\}, \quad (99)$$

or equivalently,

$$\begin{aligned}\pi_{ij} &= \exp\{-1 - \lambda d_{ij} - \lambda \alpha_i - \lambda \beta_j\} \\ &= \exp\{-1/2 - \lambda \alpha_i\} \exp\{-\lambda d_{ij}\} \exp\{-1/2 - \lambda \beta_j\}.\end{aligned}\quad (100)$$

That is, setting $x_i = \exp\{-1/2 - \lambda \alpha_i\}$, $y_j = \exp\{-1/2 - \lambda \beta_j\}$, and letting $\mathbf{A}_\lambda = (a_{ij})$ be the matrix with entries $a_{ij} = \exp\{-\lambda d_{ij}\}$, the solution $\boldsymbol{\Pi}$ is a matrix of the form

$$\boldsymbol{\Pi} = \text{diag}(\mathbf{x}) \mathbf{A}_\lambda \text{diag}(\mathbf{y}). \quad (101)$$

Because the solution $\boldsymbol{\Pi}$ must satisfy $\boldsymbol{\Pi} \mathbf{1} = \mathbf{p}$ and $\boldsymbol{\Pi}^T \mathbf{1} = \mathbf{q}$, Sinkhorn's Theorem ensures that there is a unique solution $\boldsymbol{\Pi}$ which may be obtained by applying the Sinkhorn algorithm to the matrix $\mathbf{A}_\lambda$. This algorithm has cost $O(T_\lambda \cdot n^2)$, where T_λ is the number of iterations of Sinkhorn's algorithm until convergence to a desired precision; this number typically grows with λ .

3.1 Sinkhorn's algorithm and diffusion maps

Suppose $\mathbf{x}_1, \dots, \mathbf{x}_n$ are n data vectors in $\mathbb{R}^p$. The diffusion maps algorithm begins by forming the Gaussian affinity matrix $\mathbf{W} = (w_{ij})$ with entries

$$w_{ij} = e^{-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / \epsilon}, \quad i \neq j \quad (102)$$

and $w_{ii} = 0$ for all $i = 1, \dots, n$.

In one version of diffusion maps, one normalizes $\mathbf{W}$ to be *bistochastic*, meaning that each row *and* each column sums to 1. This can be accomplished by using Sinkhorn's algorithm. The asymptotic performance of this approach has been studied in [4]. Note that the uniqueness result from Sinkhorn's theorem guarantees that the resulting matrix is still symmetric; indeed, this normalization is appealing because it preserves the symmetry of the weight matrix.

There is another advantage of this normalization, which is described in [3]. Suppose that instead of observing $\mathbf{x}_1, \dots, \mathbf{x}_n$ directly, our data is of the form $\mathbf{y}_j = \mathbf{x}_j + \mathbf{g}_j$, where $\mathbf{g}_1, \dots, \mathbf{g}_n$ are independent (but not necessarily identically distribution) vectors, with distributions $\mathbf{g}_j \sim \mathcal{N}(\mathbf{0}, (\sigma_j^2/p) \mathbf{I}_p)$ for constants σ_j .

Let $\mathbf{W}^{(y)} = (w_{ij}^{(y)})$ be a Gaussian kernel, as in diffusion maps:

$$w_{ij}^{(y)} = e^{-\|\mathbf{y}_i - \mathbf{y}_j\|^2 / \epsilon}. \quad (103)$$

On the other hand, let $\mathbf{W}^{(x)} = (w_{ij}^{(x)})$ be the idealized kernel between the signal components alone:

$$w_{ij}^{(x)} = e^{-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / \epsilon}. \quad (104)$$

Let $\mathbf{\Omega}^{(x)} = \text{diag}(\omega_1^{(x)}, \dots, \omega_n^{(x)})$ be the Sinkhorn matrix, defined so that $\mathbf{A}^{(x)} \equiv \mathbf{\Omega}^{(x)} \mathbf{W}^{(x)} \mathbf{\Omega}^{(x)}$ is bistochastic. We would like to work with $\mathbf{A}^{(x)}$, but of course, since we don't observe $\mathbf{W}^{(x)}$, we can't evaluate $\mathbf{A}^{(x)}$.

It turns out, however, that we can consistently estimate $\mathbf{A}^{(x)}$ when $p \gg n$. Take $\mathbf{\Omega}^{(y)} = (\omega_1^{(y)}, \dots, \omega_n^{(y)})$ so that $\mathbf{A}^{(y)} \equiv \mathbf{\Omega}^{(y)} \mathbf{W}^{(y)} \mathbf{\Omega}^{(y)}$ is bistochastic. It turns out that if $p \rightarrow \infty$ and n is fixed, $\mathbf{A}^{(y)}$ converges to $\mathbf{A}^{(x)}$. To see why this should be true, observe that we can write

$$\begin{aligned} \|\mathbf{y}_i - \mathbf{y}_j\|^2 &= \|\mathbf{x}_i + \mathbf{g}_i - \mathbf{x}_j - \mathbf{g}_j\|^2 \\ &= \|\mathbf{x}_i - \mathbf{x}_j\|^2 + \|\mathbf{g}_i\|^2 + \|\mathbf{g}_j\|^2 - 2\langle \mathbf{g}_i, \mathbf{g}_j \rangle - 2\langle \mathbf{x}_i - \mathbf{x}_j, \mathbf{g}_i - \mathbf{g}_j \rangle, \end{aligned} \quad (105)$$

and that as $p \rightarrow \infty$, the cross terms $2\langle \mathbf{g}_i, \mathbf{g}_j \rangle - 2\langle \mathbf{x}_i - \mathbf{x}_j, \mathbf{g}_i - \mathbf{g}_j \rangle$ all go to 0 almost surely; whereas $\|\mathbf{g}_i\|^2 + \|\mathbf{g}_j\|^2 \sim \sigma_i^2 + \sigma_j^2$; consequently, for $i \neq j$,

$$e^{-\|\mathbf{y}_i - \mathbf{y}_j\|^2 / \epsilon} \sim e^{-\sigma_i^2} e^{-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / \epsilon} e^{-\sigma_j^2}. \quad (106)$$

Consequently, if the weights $\omega_1^{(y)}, \dots, \omega_n^{(y)}$ are chosen so that the matrix with entries $\omega_i^{(y)} e^{-\sigma_i^2} e^{-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / \epsilon} e^{-\sigma_j^2} \omega_j^{(y)}$ is bistochastic, by the uniqueness of such weights they must be of the form

$$\omega_i^{(y)} = e^{\sigma_i^2} \omega_i^{(x)}. \quad (107)$$

While our derivation here is heuristic, it suggests that in the limit, the effect of the noise is completely removed by using bistochastic scaling; this is rigorously proved in [3].

References

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